

Controlling Antiferromagnetic Magnon Transport and Anisotropy by Interfacial Coupling

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Magnon spin current, which delivers spin angular momentum without charge flow, has garnered considerable interest for next-generation spintronic applications. Antiferromagnetic insulators have been shown to be the ideal material platform for magnonics. However, it remains a challenge to effectively control the antiferromagnetic magnon transport. Here, the control over both transmissivity and anisotropy of antiferromagnetic magnon transport in magnetic multilayers is reported, achieved through the delicate interplay between interfacial coupling and magnetocrystalline anisotropy. In $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ (LSMO)/ LaFeO_3 (LFO)/Pt heterostructures, a Néel vector reorientation as temperature decreases is observed through soft X-ray magnetic linear dichroism. Temperature- and angular-dependent spin pump data reveal three regions with distinct antiferromagnetic magnon transport characteristics, evolving from a magnon-conducting state with uniaxial anisotropy to an intermediate state with reduced transmissivity and weak anisotropy, and ultimately to a magnon-insulating state. Theoretical modeling reveals that this modulation is likely attributed to the evolution of both antiferromagnetic axis and domain structure of LFO, determined by the competition between exchange coupling across the interface and the intrinsic magnetic anisotropy of LFO. These findings provide fundamental insights into the critical role of the interface in controlling the magnon transport in magnetic multilayers and offer a new toolkit for developing magnonic devices.

1. Introduction

Spintronics has demonstrated remarkable potential for next-generation information technologies.^[1,2] Unlike charge current, information encoded in the spin degree of freedom can be transported in magnetically ordered insulators via collective excitation of spin dynamics, known as magnon spin current.^[3,4] The absence of net charge flow and low thermal dissipation leads to an emerging field of magnonics, holding great promise for developing energy-efficient memory and logic devices.^[5–9] Antiferromagnetic insulators, which offer advantages such as ultrafast dynamic response and negligible stray field, have been shown to support magnon spin current efficiently.^[8,10] Establishing robust control over the magnon transport in antiferromagnetic insulators is thereby highly demanded,^[11] which relies on a comprehensive understanding of the mechanisms that govern the magnon transport. Recent research endeavors have witnessed intriguing progress regarding antiferromagnetic magnon, such as long-range

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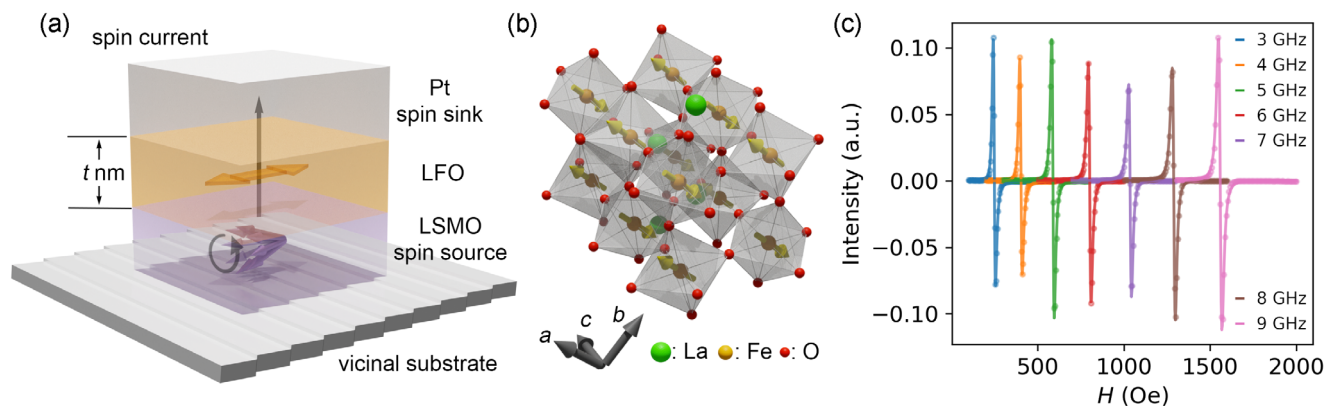


Figure 1. Sample structure for magnon transport measurements. a) Schematic illustration of spin pumping and magnon transport in the LSMO (30 nm)/LFO (*t* nm)/Pt (8 nm) heterostructure. b) Schematic illustration of LFO crystal structure and antiferromagnetic Neel vector orientation. c) Ferromagnetic resonance spectra measured at 300 K with varying frequencies.

magnon transport,^[12,13] magnon-mediated spin valve,^[14,15] magnon torques,^[7,16] and voltage-controlled magnon.^[17–20] In magnetic multilayers, a large modulation of magnon transmission has been reported in the vicinity of antiferromagnetic phase transition in Cr₂O₃, NiO, and CoO due to spin fluctuations, which can be tuned by external fields in a narrow temperature range.^[21–23] Isothermal control of magnon transport has also been achieved through a spin-flop transition induced by high magnetic fields in antiferromagnets with large canting moment and weak anisotropy.^[24] More recently, electric-field control of antiferromagnetic order in multiferroics has been exploited for energy-efficient manipulation of magnon transport.^[17,25] These approaches to control magnon transport are mainly based on manipulating the intrinsic magnetic properties within an antiferromagnetic insulator itself. The interfacial coupling between antiferromagnet and ferromagnet, a well-known effect in magnetic multilayers,^[26–28] provides another pathway to tune the magnon transport. This coupling has been shown to reverse the sign of the spin Seebeck signal,^[29] spin-Hall magnetoresistance,^[30] and spin-orbit torque.^[31] However, its influence on controlling magnon transmissivity and anisotropy, especially when closely competing with the intrinsic properties of antiferromagnets, has not yet been explicitly investigated, potentially offering a new perspective for the design of magnonic devices.

In this work, using La_{0.7}Sr_{0.3}MnO₃ (LSMO)/LaFeO₃ (LFO)/Pt heterostructures as the model system, we demonstrate that the interfacial exchange coupling provides an efficient route to control both transmissivity and anisotropy of magnon transport. The temperature- and angular-dependent X-ray magnetic linear dichroism (XMLD) spectra demonstrate that the LFO Néel vector changes from a fixed orientation to the spin-flop coupling geometry with LSMO magnetization as the temperature decreases. Spin pumping measurements reveal a sharp transition of magnon transmission upon decreasing temperature, distinct from previous reports based on antiferromagnetic phase transition. Moreover, the angular-dependent magnon transport uncovers three distinct behaviors of the magnon transport: i) a high-temperature state with a high magnon transmission and strong uniaxial anisotropy, ii) an intermediate-temperature state

with a reduced transmission and weak anisotropy, and iii) a low-temperature state with fully blocked magnon transmission. Theoretical modeling validates that the interplay between magneto-crystalline anisotropy and interfacial exchange coupling dictates the change in Néel vector orientation and antiferromagnetic domains, which governs the different magnon transport behaviors. Our findings not only deepen the fundamental understanding of magnon spin transport in magnetic multilayers but also pave the way for versatile magnonic applications.

2. Sample Structure and Antiferromagnetic Order

To study the magnon transport behaviors, heterostructures composed of 30 nm-thick LSMO, LFO with varied thickness (*t* nm), and 8 nm-thick Pt were deposited on vicinal SrTiO₃ substrates with a 4° miscut along [100] crystalline direction, as depicted in Figure 1a. The heterostructures exhibit high crystalline quality and sharp interfaces (Figure S1, Supporting Information). Bulk LFO adopts an orthorhombic perovskite structure, with its Néel vector strongly coupled to the orthorhombic *a*-axis as depicted in Figure 1b. As previously demonstrated, the vicinal substrates break the crystalline symmetry, leading to a single-domain LFO film with the Néel vector projected along the [100] direction as shown in Figure 1a.^[32–35] Figure 1c displays the ferromagnetic resonance (FMR) spectrum of the LSMO layer in the heterostructures at room temperature, exhibiting a high effective magnetization (0.36 T) and a low damping factor (0.0032). FMR measurements performed at various temperatures reveal that both the linewidth and amplitude of the FMR spectrum exhibit a relatively weak temperature dependence unless approaching the Curie temperature above room temperature (Figure S2, Supporting Information). These properties suggest that LSMO provides a good spin source that can efficiently generate spin current via spin pumping over a wide temperature range.

To determine the Néel vector orientation, we first performed soft X-ray magnetic linear dichroism (XMLD) measurements. As illustrated in Figure 2a, the incident X-ray beam was normal to the sample surface, and X-ray absorption spectroscopy (XAS) spectra were collected at Fe *L*_{2,3} absorption edges with the electric-field vector oriented both perpendicular and parallel to

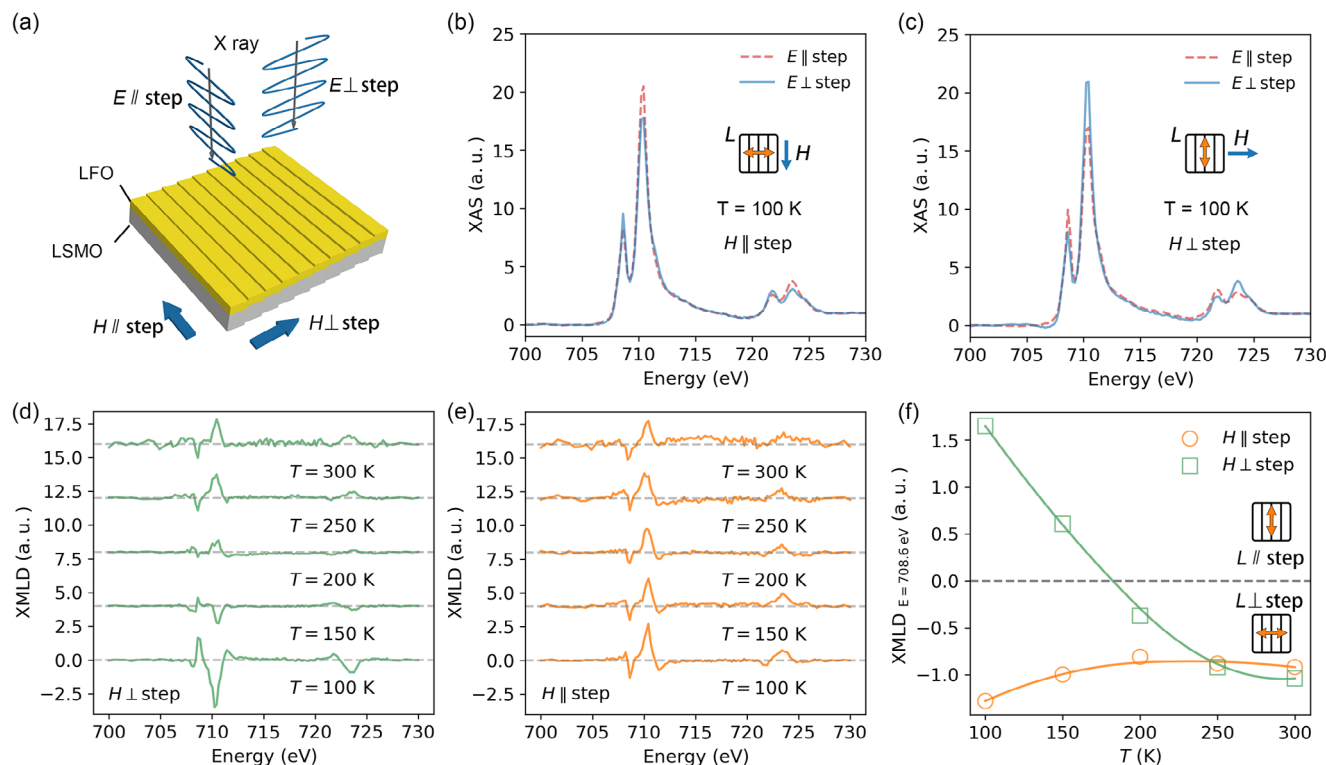


Figure 2. X-ray magnetic linear dichroism (XMLD) results. a) Schematic representation of XMLD measurement setup. The X-ray absorption spectroscopy (XAS) spectra were measured when the electric field of the X-ray was parallel and perpendicular to the substrate step edges. b,c) XAS spectrums measured at 100 K with electric-field parallel (orange dash) and perpendicular (blue solid) to the step edges, when magnetic fields with orthogonal directions were applied. The insets show the relative orientation of the Néel vector with respect to the magnetic field revealed by XMLD. d,e) Temperature evolution of XMLD spectra with magnetic field perpendicular (d) and parallel (e) to the step edges. f) The temperature-dependent XMLD intensity at a specific energy ($E = 708.6$ eV) from 100 K to 300 K with different magnetic field directions. The positive/negative sign corresponds to the Néel vector parallel/perpendicular to the step edges, as shown in the Figure inset.

the substrate step edges. The XMLD signal is defined as the difference in XAS intensities, which reveals the in-plane orientation of the LFO Néel vector. During these measurements, a 2000 Oe in-plane magnetic field was applied to saturate LSMO magnetization. Figure 2b shows the XAS spectra of Fe $L_{2,3}$ absorption edges under an applied magnetic field parallel to step edges at 100 K. Based on previous reports, the differences in XAS spectra at both L_3 and L_2 edges reveal that the LFO Néel vector is perpendicular to the step edges in this case,^[33] as schematically shown in the Figure inset. Upon rotating the magnetic field by 90 degrees at 100 K, the sign of the XMLD signal reverses, indicating that the rotation of the Néel vector to be along the step edges as shown in Figure 2c. These results reveal the spin-flop coupling geometry at low temperature,^[36] i.e., the in-plane projection of the LFO Néel vector is orthogonally coupled to LSMO magnetization. Note that this Néel vector rotation is absent in the LFO single layer when subjected to the same magnetic fields (Figure S3, Supporting Information), confirming the crucial role of the interface.

Figure 2d,e shows the XMLD spectra measured over a broad temperature range (100–300 K), when the magnetic field is perpendicular to (Figure 2d) or parallel to (Figure 2e) the substrate step edges, respectively. The sign of XMLD signals remains unchanged at 300 K, revealing that the Néel vector aligns perpendicular to step edges regardless of magnetic field direc-

tion. This is consistent with the intrinsic magnetic anisotropy of the orthorhombic LFO. Figure 2f summarizes the temperature-dependent XMLD signals at a fixed photon energy ($E = 708.6$ eV), where a positive (negative) sign corresponds to the Néel vector parallel (perpendicular) to the step edges. A pronounced sign reversal of the XMLD signal upon cooling is observed when the magnetic field is perpendicular to the step edges ($H \perp$ step), but is absent for the other magnetic field direction ($H \parallel$ step). These results demonstrate the transition from the high-temperature state dominated by the intrinsic magnetic anisotropy of LFO to the low-temperature state dominated by the spin-flop interfacial coupling. To further reveal the antiferromagnetic order, we also measured the angle-dependent XAS by rotating the electric-field vector of the incident X-ray (Figure S4, Supporting Information). The angular dependent data at 300 and 100 K verify the Néel vector orientation as established above. Moreover, the results at 180 K exhibit no angular dependence, suggesting the formation of multiple antiferromagnetic domains.

3. Magnon Conductor–Insulator Transition

Spin pumping measurements were employed to characterize the magnon transport.^[37] As schematically illustrated in Figure 3a, a 4 GHz microwave was applied via a coplanar waveguide (CPW), which induces coherent magnetization precession

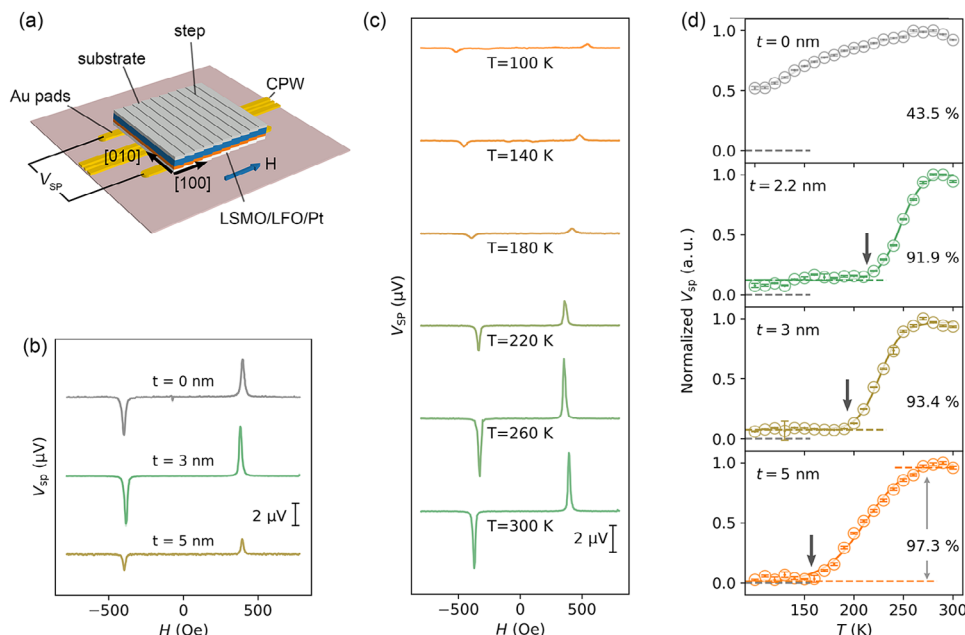


Figure 3. Temperature-dependent spin pumping signals. a) Schematic representation of spin pumping measurement setup. b) Spin pumping voltage (V_{SP}) as a function of external magnetic field for heterostructures with different LFO thickness (t) at 300 K. c) Magnetic field-dependent spin pumping voltage (V_{SP}) measured in the heterostructure with 3 nm-thick LFO from 100 to 300 K. d) Temperature dependence of the magnon transmission for samples with different LFO thicknesses. The error bars represent standard deviations arising from uncertainties in the fits of the spin pumping voltage amplitudes versus the magnetic fields.

within the LSMO layer and generates a spin accumulation at the LSMO/LFO interface. The magnon current with spin polarization oriented along the LSMO magnetization is then transmitted through the LFO layer and ultimately reaches the Pt layer, where it is converted into a measurable voltage (V_{SP}) through the inverse spin Hall effect.^[38]

The magnetic field-dependent spin pumping voltage can be fitted by a Lorentzian curve:

$$V_{SP} = V_{ISHE} \frac{\Delta H^2}{\Delta H^2 + (H - H_{res})^2} + V_{AHE} \frac{\Delta H (H - H_{res})}{\Delta H^2 + (H - H_{res})^2} \quad (1)$$

where the V_{ISHE} and V_{AHE} are the voltage of inverse spin pumping and anomalous Hall effect, ΔH is the linewidth of the Lorentzian curve, and H_{res} is the resonance field. The V_{AHE} is negligible for all samples. The thickness-dependent magnon transport at 300 K is shown in Figure 3b for various LFO thicknesses, with the magnetic field applied perpendicular to the step-edges of the vicinal substrate ([100] direction). For thinner LFO layers, a robust spin pumping signal is evident, which reduces as the LFO thickness increases beyond 5 nm. This attenuation underscores the role of LFO thickness in modulating the magnon diffusion. Further insights emerge from the temperature-dependent data in Figure 3c for a heterostructure with a 3 nm thick LFO. At 300 K, the spin pumping signal exhibits a pronounced resonance peak ≈ 400 Oe. Reversing the applied magnetic field polarity inverts the sign of V_{SP} , confirming its origin from spin pumping. As the temperature decreases, the resonance peak amplitude largely reduces, revealing the transition from magnon-conducting to magnon-insulating state.

Figure 3d compares the temperature dependence of the normalized V_{ISHE} signals in LSMO/Pt bilayers ($t = 0$ nm) and LSMO/LFO/Pt trilayers with varying LFO thicknesses ($t = 2.2, 3, 5$ nm). The LSMO/Pt bilayer only exhibits a modest decrease in V_{ISHE} upon cooling, consistent with previous reports.^[39] In contrast, the trilayer with 2.2 nm LFO shows an abrupt decline near 250 K, indicating that this rapid magnon conducting-insulating transition is correlated with the presence of the LFO layer rather than the change in LSMO or Pt properties. By increasing the LFO thickness, we find that the leakage of magnon spin current at low temperature decreases to nearly zero, leading to a nearly 100% modulation ratio as defined by $(V_{ISHE}(300 \text{ K}) - V_{ISHE}(100 \text{ K})) / V_{ISHE}(300 \text{ K})$. Moreover, the decrease of V_{ISHE} becomes more gradual, and the temperature at which V_{ISHE} diminishes (marked by the arrow) shifts to the lower temperature side as increasing LFO thickness increases. Note that similar modulation of V_{ISHE} by decreasing temperature has been reported in similar trilayer structures with antiferromagnetic insulators such as Cr_2O_3 ^[21] or CoO ,^[23] which is attributed to the antiferromagnetic phase transition. In those reports, the magnon conducting-insulating transition shifts to higher temperatures with increasing antiferromagnet thickness, which is in sharp contrast to our observation, suggesting a different mechanism governing magnon transport in our heterostructures.

4. Anisotropic Magnon Spin Transport

By rotating the sample, we conducted angular-dependent spin pumping measurements at various temperatures. The angle between the [100] direction and magnetic field (along CPW) is defined as ϕ (Figure 4a). Since the spin polarization of the

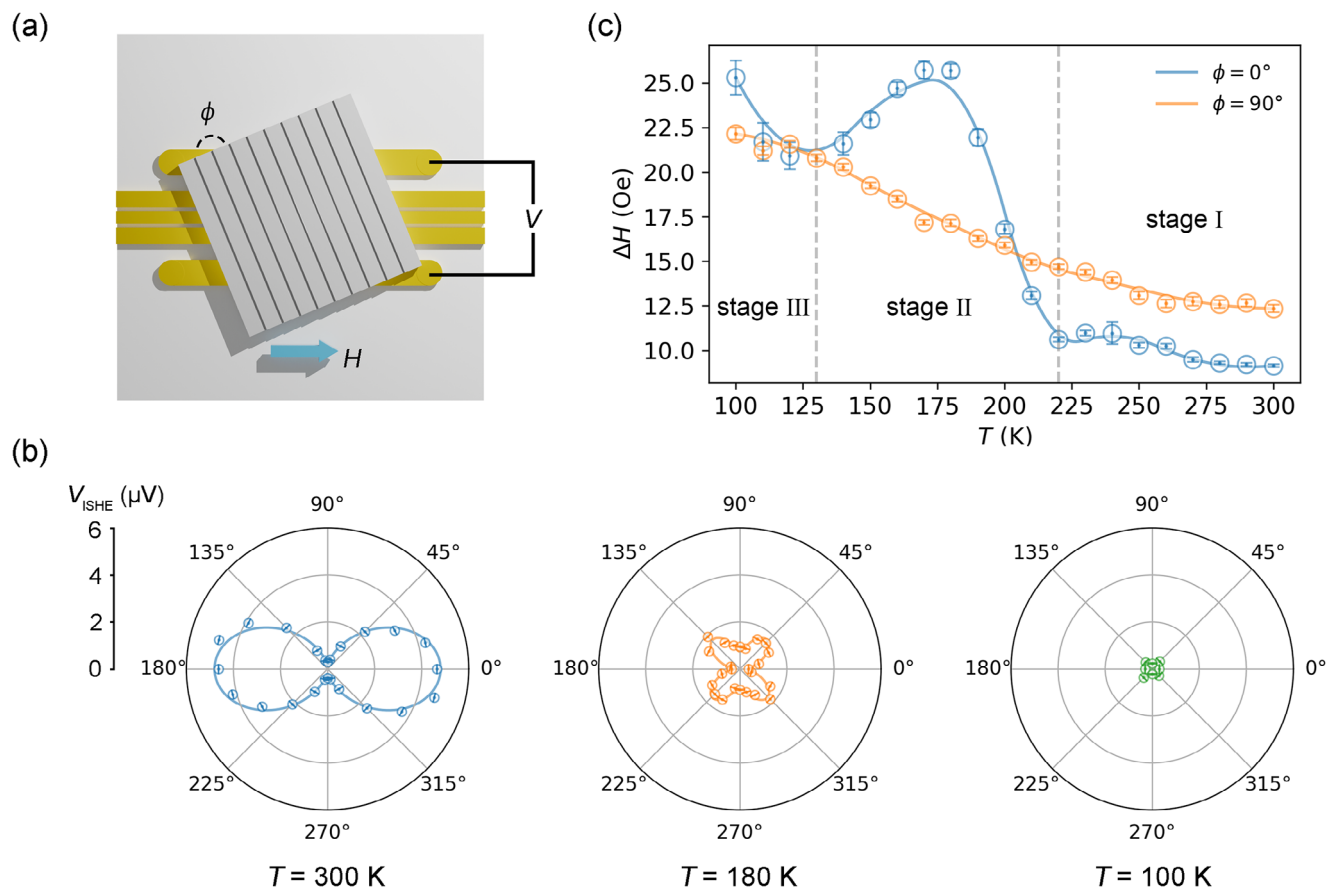


Figure 4. Angular-dependent spin pumping signals. a) Schematic representation of measurement setup, showing that the sample is rotated by an angle ϕ with respect to the magnetic field (and the coplanar waveguide). b) Angular (ϕ) dependence of inverse spin Hall voltage (V_{ISHE}) measured at 300, 180, and 100 K for the LSMO/LFO/Pt with 3 nm LFO. c) Temperature-dependent linewidths of the spin pumping signals for $\phi = 0^\circ$ and 90° .

injected spin current aligns with the applied magnetic field, this experimental setup allows us to investigate the magnon transport anisotropy depending on the spin polarization direction. Figure 4b shows the angular dependence of the inverse spin Hall voltage (V_{ISHE}) for the sample with $t = 3$ nm at three representative temperatures of 300, 180, and 100 K, exhibiting distinct magnon transport behaviors. At 300 K, the trilayer exhibits a high magnon transmission with uniaxial anisotropy. Magnon spin current can efficiently transmit through the LFO layer at $\phi = 0^\circ$ and 180° where the injected spin polarization aligns with the Néel vector, but is fully blocked at $\phi = 90^\circ$ and 270° , leading to a pronounced two-fold rotation symmetry of magnon transport. As the temperature decreases to an intermediate range (such as 180 K), the magnon transmission largely reduces and changes to a nearly four-fold rotation symmetry. Note that the LSMO/Pt bilayer also exhibits similar four-fold rotation symmetry, likely associated with the intrinsic magnetic anisotropy of LSMO (Figures S5 and S6, Supporting Information). This indicates a relatively weak anisotropy of magnon transport across LFO in this temperature range (Figure S6, Supporting Information). We also find that the V_{ISHE} drops faster at $\phi = 0^\circ$ as compared to other ϕ angles. Finally, at 100 K, the LFO layer almost completely impedes the magnon transport regardless of the spin polarization direction of the injected spin current.

Figure 4c shows the temperature-dependent linewidth (ΔH) of spin-pumping signals at $\phi = 0^\circ$ and 90° , which also reveals three distinct regimes. In stages I (above 220 K), the linewidth at both $\phi = 0^\circ$ and 90° gradually increases as temperature decreases. In stage II (≈ 220 – 130 K), a pronounced hump appears in the linewidth at $\phi = 0^\circ$ but not at $\phi = 90^\circ$, indicating an unusual spin-damping mechanism. Last, in stage III below 130 K, the linewidth values for $\phi = 0^\circ$ and 90° converge and further increase as the temperature decreases. Note that this non-monotonic change of ΔH is not observed in LSMO/Pt bilayers and LSMO single layer (Figures S2 and S7, Supporting Information). Since the ΔH is closely related to the spin dynamics of LSMO, which is affected by the interface, these results suggest variations of exchange coupling between LSMO and LFO as temperature decreases, which are likely correlated to the different magnon transport behaviors shown in Figure 4b.

5. Mechanism for Controlling Magnon Transport

To shed light on the underlying mechanism, we investigate the evolution of Néel vector and magnetic domain structure by using the micro-spin simulation, which considers both the magneto-crystalline anisotropy (MAE , $K_{\text{AFM}}t_{\text{AFM}}$) of LFO and the interfacial magnetic coupling (J_{I}) at the LSMO/LFO interface (see

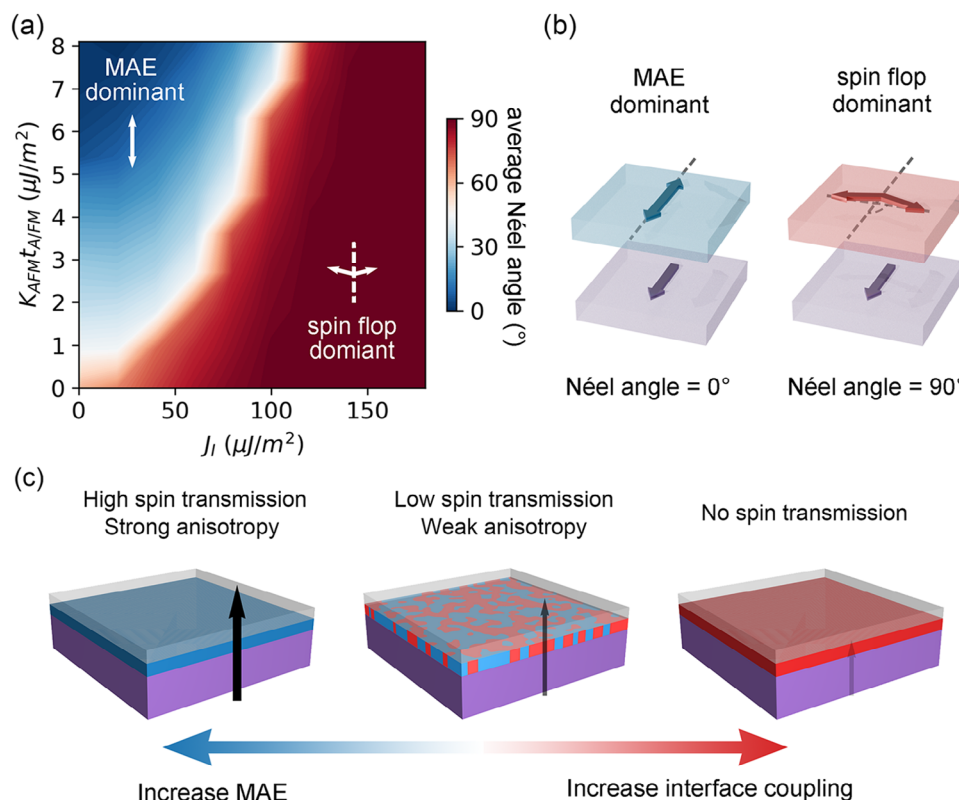


Figure 5. Mechanism for interface-controlled magnon transport. a) Simulation results on the evolution of Néel vector orientation by tuning the competition between magneto-crystalline anisotropy (MAE) and interfacial magnetic coupling. b) Schematic illustrations of the Néel vector orientation in two extreme cases. c) Schematic illustrations of the antiferromagnetic states by varying MAE and interfacial coupling, leading to three distinct regions with tunable magnon transmission and anisotropy. The red/blue regions represent the antiferromagnetic domains of LFO with Néel vector orientation as shown in (b).

Method Note S1, Supporting Information for details).^[40] Figure 5a shows the energetically favorable azimuthal angles of LFO moments as a function of those two energies, assuming the LSMO magnetization aligns parallel to the intrinsic antiferromagnetic axis of LFO (i.e., perpendicular to the step edges). When the magneto-crystalline anisotropy energy dominates, the Néel vector of LFO remains stable along its intrinsic easy axis with an almost single-domain state (left panel of Figure 5b; Figure S4, Supporting Information), determined by the orthorhombic symmetry. In this case, magnons can efficiently propagate through LFO when the spin polarization aligns with the Néel vector, leading to a high magnon transmission with strong uniaxial anisotropy at high temperatures (left panel of Figure 5c). In contrast, when the interfacial coupling becomes dominant, the orthogonal coupling of the Néel vector and LSMO magnetization is energetically favorable (right panel of Figure 5b), which leads to the full blockage of magnon transport at low temperature (right panel of Figure 5c). Intriguingly, when the two energy terms are comparable in magnitude, our simulation results reveal the presence of multiple antiferromagnetic domains with different Néel vectors for the LFO layer (Figure S8, Supporting Information). Note that this is consistent with the absence of XMLD signal by varying the electric-field vector of incident X-ray (Figure 2; Figure S4, Supporting Information) and the anomalous enhancement of linewidth (Figure 4b) observed in the intermediate tempera-

ture range. Owing to the multiple antiferromagnetic domains, the magnon transport shows a reduced transmission and relatively weak anisotropy (middle panel of Figure 5c).

Our model not only accounts for the temperature-dependent evolution of magnon transport but also explains the thickness-dependent behavior of V_{ISHE} as shown in Figure 3d. Since the magneto-crystalline anisotropy energy of LFO is proportional to its thickness (i.e., $K_{\text{AFM}}t_{\text{AFM}}$), the interfacial coupling can more effectively rotate the Néel vector for thinner LFO, leading to the sharp magnon conducting-insulating transition with a high transition temperature. In contrast, as the LFO thickness increases (larger $K_{\text{AFM}}t_{\text{AFM}}$), a larger interfacial coupling strength is required to induce the spin-flop coupling, leading to a shift of the transition toward lower temperatures, as shown in Figure 3d.

6. Conclusion

In conclusion, our work demonstrates a strategy to control antiferromagnetic magnon transport in magnetic multilayers by exploiting the interplay between interfacial magnetic coupling and intrinsic magnetic anisotropy of antiferromagnets. In LSMO/LFO/Pt heterostructures, a magnon conducting-to-insulating transition and a pronounced change in its anisotropy are observed as temperature decreases, achieved through modulating the antiferromagnetic axis and domain structure. This

interface-driven mechanism is generic and can be extended to magnetic heterostructures with a wide range of materials. Recently, the electric-field control of magnon transport has attracted emerging research interests for energy-efficient magnonic devices.^[17,19,41–44] Our findings suggest that the modulation of interfacial coupling by electric field, as established in multiferroic heterostructures,^[45–47] could provide another promising route to develop a highly efficient magnon switch.

7. Experimental Section

Sample Growth: $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3/\text{LaFeO}_3$ heterostructures were epitaxially deposited on vicinal SrTiO_3 [001] substrates by using KrF excimer pulsed laser deposition (PLD). The sequential deposition of $\text{La}_{0.7}\text{Sr}_{0.3}\text{MnO}_3$ and LaFeO_3 layers was carried out under different oxygen partial pressures, 25 Pa for LSMO and 6 Pa for LFO. Throughout the deposition, the temperature was maintained at 780 °C. After the deposition, the samples were gradually cooled down to ambient temperature at a cooling rate of 15 °C min^{−1}, under an oxygen atmosphere of 1000 Pa. An 8 nm layer of Pt was subsequently deposited in situ on the heterostructure through DC sputtering, under an Argon atmosphere of 0.3 Pa, following chamber re-evacuation.

XMLD Measurement: Soft X-ray magnetic linear dichroism (XMLD) spectroscopy experiments were carried out using total electron yield (TEY) at beamline 07U at Shanghai Synchrotron Radiation Facility (SSRF) at varying temperatures. The X-ray energy was varied between 690 and 730 eV to cover the Fe $L_{2,3}$ edges. The incident angle of the X-ray was 90° to the sample surface, with the electric field component parallel and perpendicular to the steps. The angular-dependent X-ray absorption spectroscopy (XAS) measurements with varying electric-field vector orientations were carried out at Branch I10-1 of Diamond Light Source.

Ferromagnetic Resonance Measurements: The ferromagnetic resonance spectra were measured using the CryoFMR-40 spectrometer developed by NanOsc Instruments, integrated within the Physical Property Measurement System (PPMS) from Quantum Design. An alternating current (AC) magnetic field at 490 Hz was generated by a Helmholtz coil, and the derivative of the transmission parameter S_{21} (dI/dH) was measured.

Spin Pumping Measurement: The assessment of spin pumping (SP) was executed within the chamber of the Physical Properties Measurement System (PPMS, Quantum Design). The ferromagnetic resonance of LSMO was stimulated by utilizing a microwave emission, characterized by a 20 dBm power output and a 4 GHz frequency, through a coplanar waveguide arrangement. This microwave emission was modulated by an alternating current voltage of 1 Vrms at a frequency of 1713 Hz. The voltage was measured through the SR830 lock-in amplifier.

Magnetic Structure Simulation: As for micro-spin simulation, the anti-ferromagnetic exchange $\delta_{\text{AFM}} = 1.25 \times 10^5 \text{ J m}^{-3}$ was used. The anisotropy energy K_{AFM} was varied from 0 to 3 kJ m^{−3}, and interfacial coupling energy J_i was varied from 0 to 0.2 mJ m^{−2}, based on the parameters used in previous reports for LFO/LSMO bilayers.^[40] This enables us to reveal how the competition of the two parameters affects the Néel vector of LFO. The thickness (t_{AFM}) was set to be 3 nm according to the experiments. The simulation was performed on a 64 × 64 grid with a cell size of 8 nm. The stable state was obtained by locating the minimum energy using a gradient descent algorithm.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

antiferromagnets, interface coupling, magnon current, spin pumping

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